Brian Huang

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Education

Boston University | College of Engineering — Boston, MA
Bachelor of Science in Computer Engineering — *Sep. 2025 – (Expected) May 2029*

Technical Skills

Languages: Python, C/C++, TypeScript, Java, JavaScript, SQL, HTML/CSS, R
Frameworks & Tools: React, Vite, Tailwind CSS, Git/GitHub, Vercel, ROS 2, Gazebo
Hardware: PC Assembly (Troubleshooting, Overclocking, & Repair), SOLIDWORKS, 3D Printing, AV Systems
AI/ML: Claude Code & API, MCP, Ollama/llama.cpp, LLM Integration, Tool Calling, Embeddings, RAG

Experience

Lead Software Engineer — Oct. 2026 – Present
GradientV LLC | Technology & Systems Integration Agency — *New York, NY*
- Led engineering at an AI and custom software agency serving 3+ paying small-business clients.
- Built and deployed 3 production client websites on Vercel for a tax advisory firm, a Filipino cafe, and a food truck, with custom domains, SEO and social-share metadata, and accessible, mobile-responsive layouts.
- Built a self-assessment tool that estimates IRS penalty exposure by industry, plus catering and signup forms.
- Developed AI tooling (LLM agents, RAG, API integrations) that lets owners query their own business data.

Student TechOps Assistant — Sep. 2026 – Present
Boston University | College of Communication — *Boston, MA*
- Ran evening operations for COM's broadcast studios, podcast suites, editing bay, and classrooms.
- Troubleshot classroom AV, Zoom, and macOS editing workstations running Adobe Creative Cloud.
- Provided customer-facing tech support to students and faculty as the evening point of contact.

Founder & Website Developer — Jan. 2023 – Present
Cubic Builds | Custom PC Building Business — *New York, NY*
- Founded a custom PC building business, selling 10+ systems at a $1,000 average for $10K+ revenue.
- Tailored each build to the client's budget and use case (gaming, streaming), balancing speed and aesthetics.
- Sourced, built, and stress-tested each PC (Prime95, MemTest86, OCCT, FurMark) for clients across the US.

Projects

BU Dining Hall Meal Planner — bumeals.vercel.app
Full-Stack Developer (TypeScript, React, Tailwind CSS, Vite, Node.js, Cheerio, Vercel) — *Sep. 2026 – Present*
- Built a web app that turns live menus from BU's 4 dining halls into macro-based meal plans.
- Wrote scrapers for BU Dining and Sodexo menus, auto-refreshed twice daily with GitHub Actions.
- Designed a meal-planning algorithm averaging 4% calorie error across 92 automated test scenarios.
- Integrated the Claude API so students can adjust their plans in plain English without fake menu items.

BU AI Society | Local LLM Division — Website
LLM Inference Researcher (Ollama/llama.cpp, Python) — *Sep. 2026 – Present*
- Profiled 8-bit Qwen3-1.7B in llama.cpp, measuring prefill vs. decode speed and TTFT on 50 to 4K-token prompts.
- Presented a live demo to a 20+ member group on why LLM text generation is memory-bandwidth-bound.
- Plotted throughput and TTFT against prompt length, showing prefill ran [N]x faster per token than decode.

BU AI Society | Robotics Division (Project JetHexa) — Website
Robotics Simulation Developer (ROS 2, Gazebo, WSL/Docker) — *Sep. 2026 – Present*
- Built a ROS 2 and Gazebo simulation of an 18-DOF hexapod robot with lidar, RGB-D camera, and IMU.
- Validated the sim against hardware specs: servo limits, forward kinematics, and URDF-to-SDF sensors.
- Documented reproducible Docker, WSL, and VM setups for an 8-person team training a PPO walking gait.

BU Rocket Propulsion Group | Project HIP — Website
Nozzle Design Engineer (SOLIDWORKS, MATLAB, Python) — *Jan. 2026 – Present*
- Designed nozzle geometry for a student-built hybrid rocket engine in SOLIDWORKS, sized for target altitude.
- Refined nozzle geometry and materials through team design reviews ahead of fabrication and testing.
- Integrated nozzle and electrical systems with the team toward engine fabrication and a static-fire test.